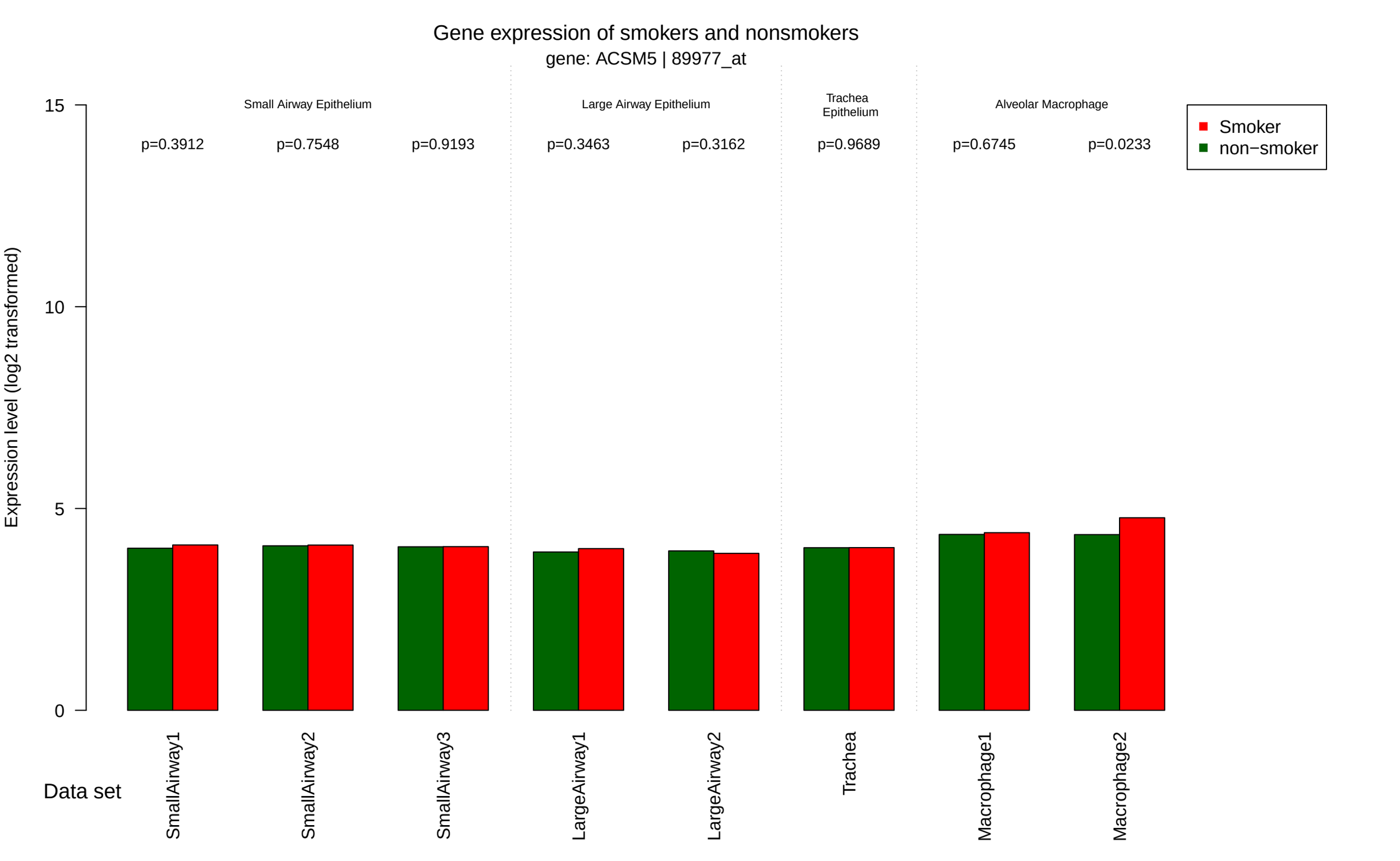
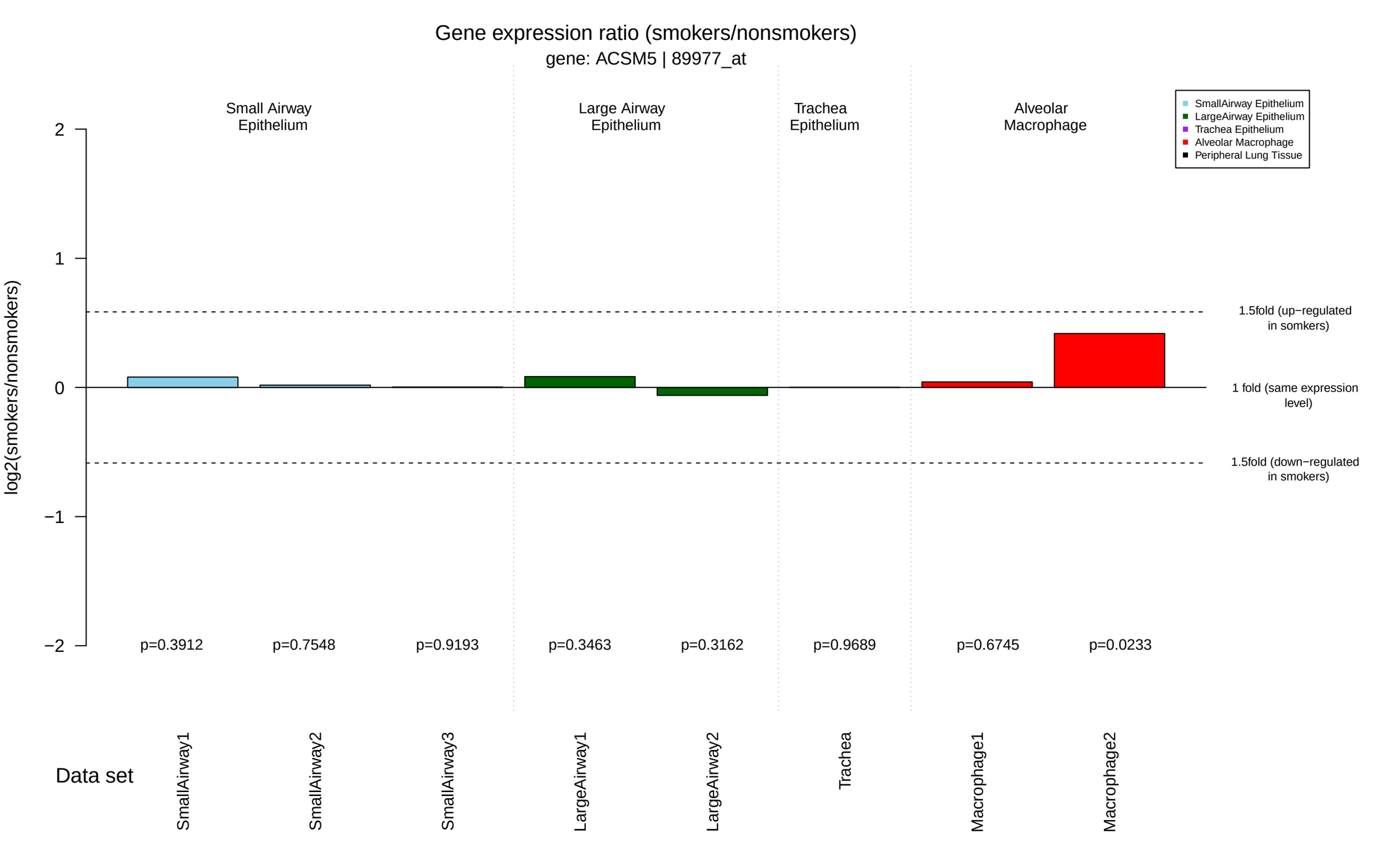
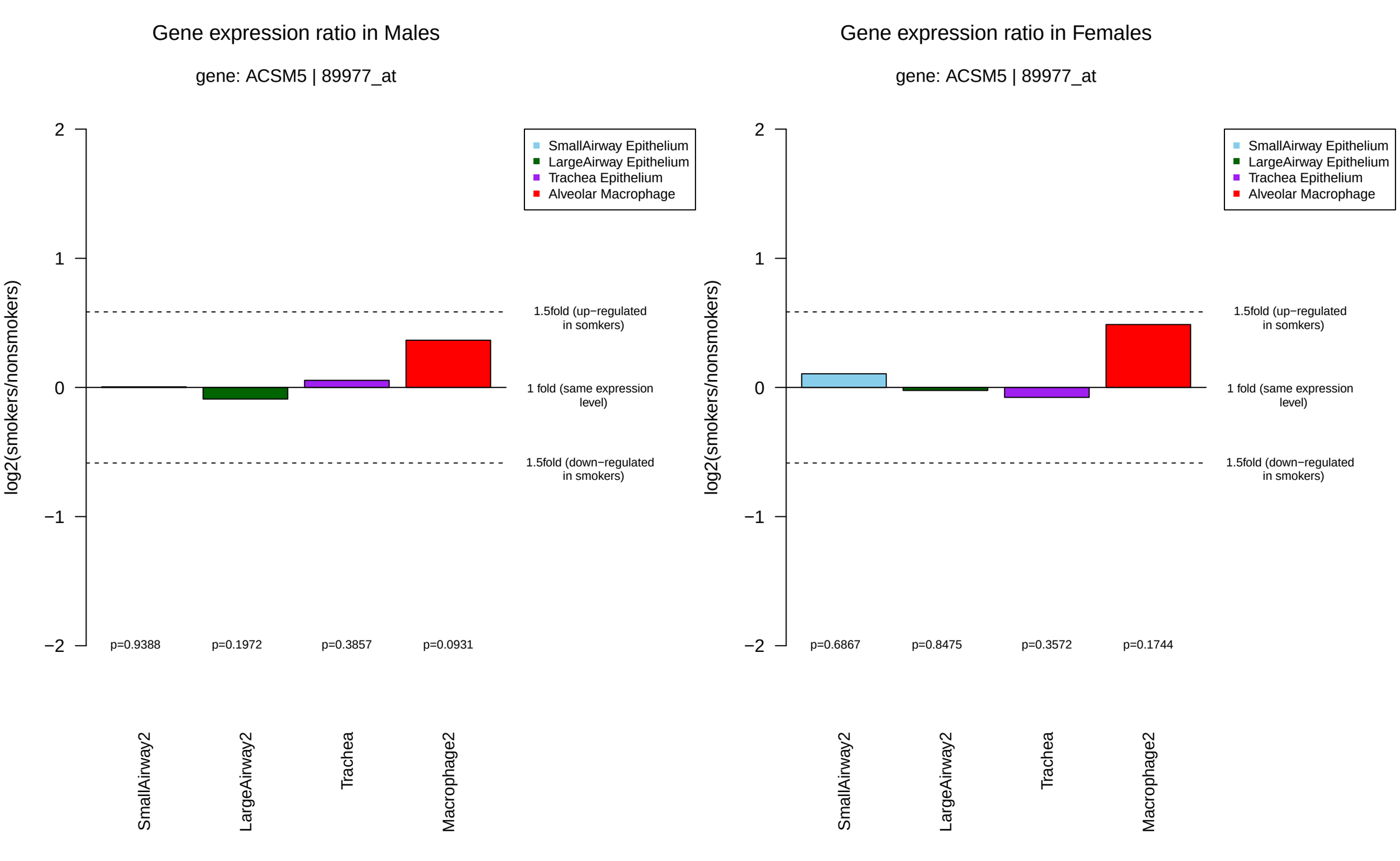
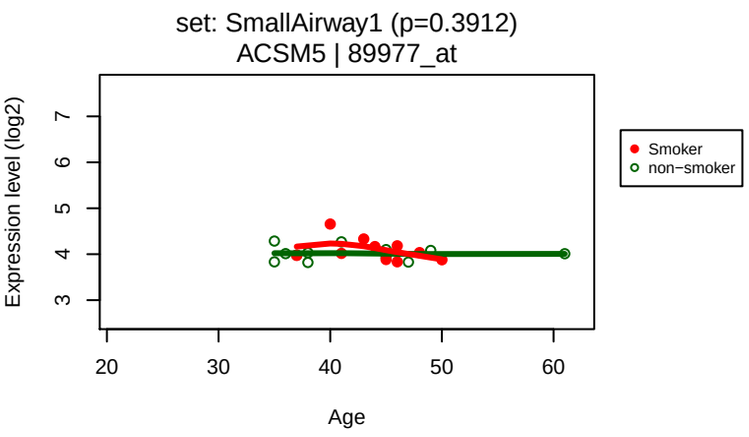
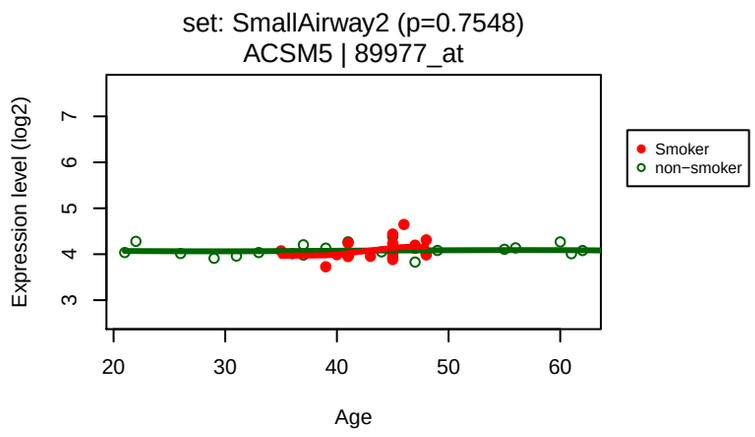
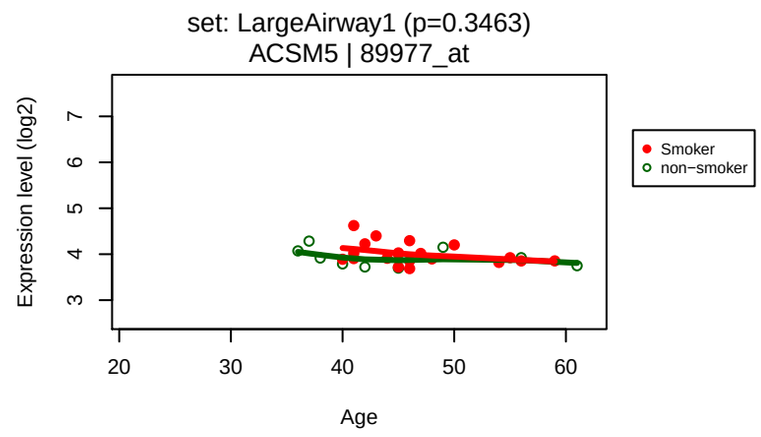

Gene expression of smokers and nonsmokers
gene: ACSM5 | 89977_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3912
p=0.7548
p=0.9193
p=0.3463
p=0.3162
p=0.9689
p=0.6745
p=0.0233
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

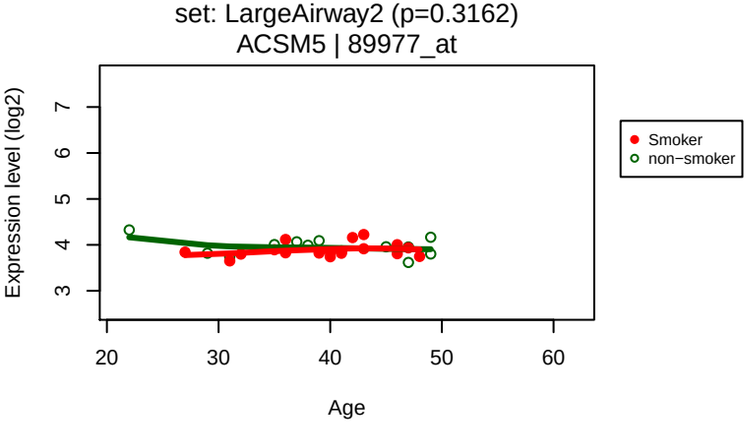

Gene expression ratio (smokers/nonsmokers)
gene: ACSM5 | 89977_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3912
p=0.7548
p=0.9193
p=0.3463
p=0.3162
p=0.9689
p=0.6745
p=0.0233
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

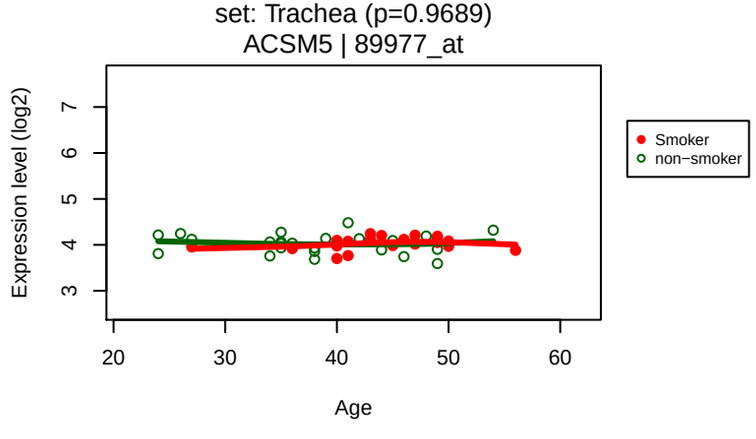

Gene expression ratio in Males
gene: ACSM5 | 89977_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9388
p=0.1972
p=0.3857
p=0.0931
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACSM5 | 89977_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6867
p=0.8475
p=0.3572
p=0.1744
SmallAirway2
LargeAirway2
Trachea
Macrophage2

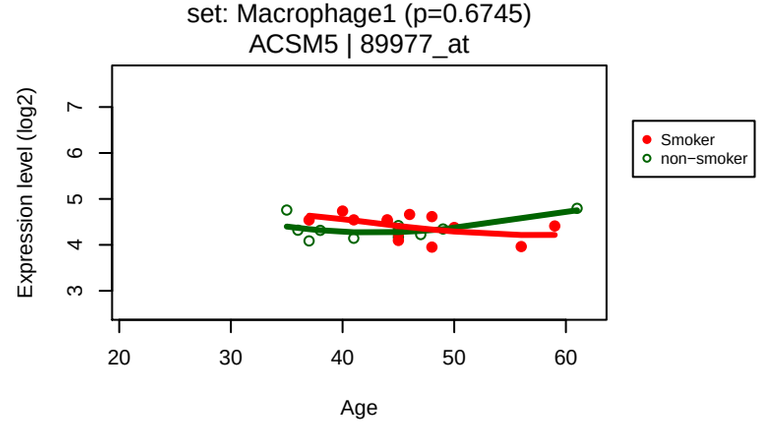

set: SmallAirway1 (p=0.3912)
ACSM5 | 89977_at
Expression level (log2)
Age
Smoker
non−smoker

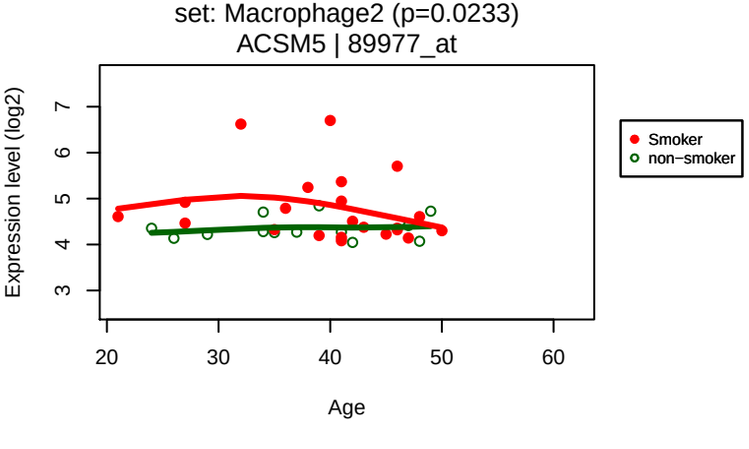

set: SmallAirway2 (p=0.7548)
ACSM5 | 89977_at
Expression level (log2)
Age
Smoker
non−smoker

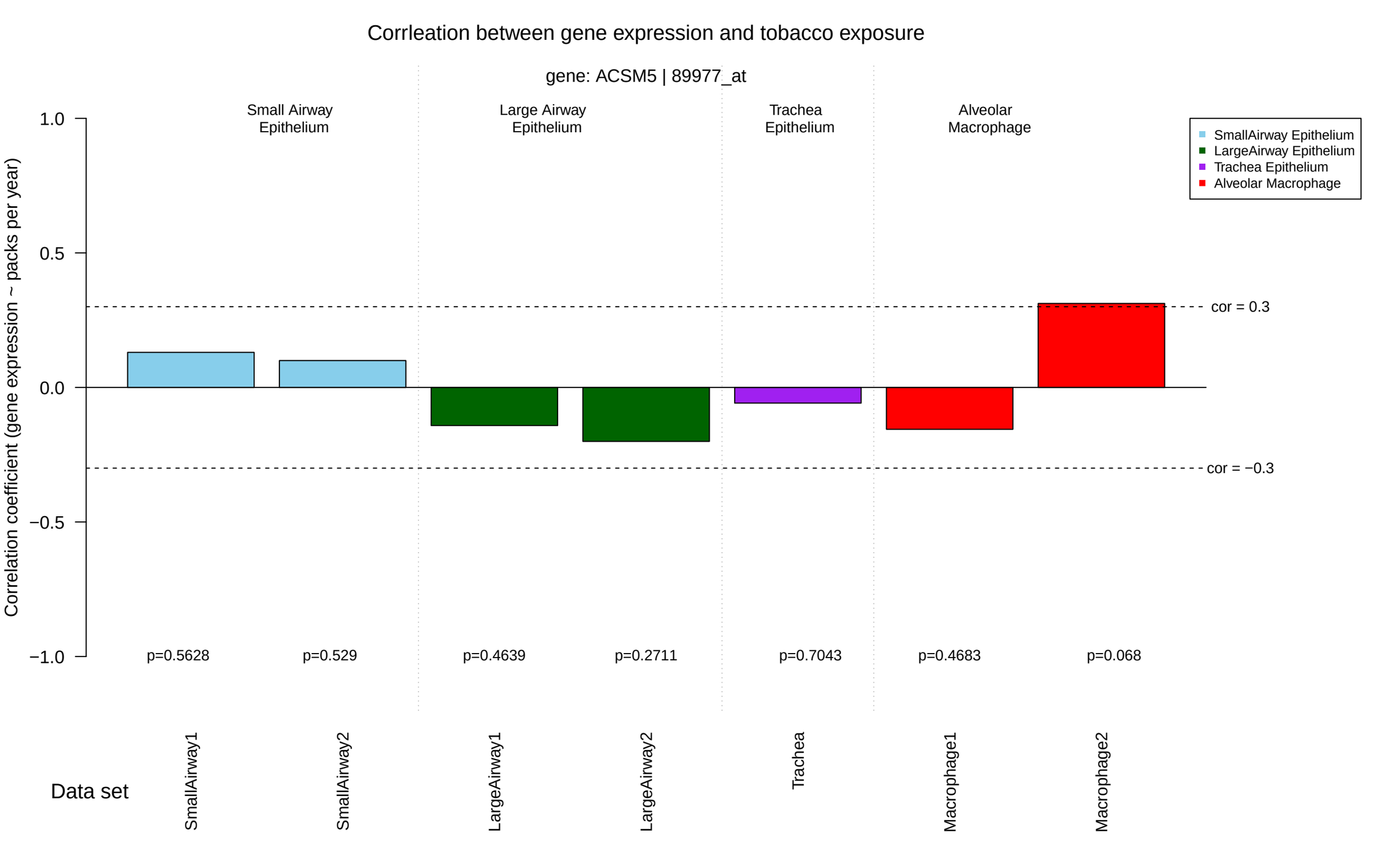

set: LargeAirway1 (p=0.3463)
ACSM5 | 89977_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3162)
ACSM5 | 89977_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9689)
ACSM5 | 89977_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6745)
ACSM5 | 89977_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0233)
ACSM5 | 89977_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACSM5 | 89977_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5628
p=0.529
p=0.4639
p=0.2711
p=0.7043
p=0.4683
p=0.068
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2